David Nicholson

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Summary

Data Scientist experienced in architecting machine learning workflows, fine-tuning LLMs, and managing multi-tenant database systems. Adept at bridging technical data engineering with business decision-making through interactive dashboards, automated metadata pipelines, and production-ready NLP models.

Skills & Proficiency

- **Machine Learning & NLP**: PyTorch, TensorFlow, Scikit-Learn, Transformers, Large Language Models (Claude, Gemini, Ollama), Clustering (HDBSCAN, K-Means), Dimensionality Reduction (UMAP), Document Embeddings, Knowledge Graphs, Entity Resolution, Text Mining and Extraction, Automatic Metadata Extraction, Weak Supervision (Snorkel)
- **Data Engineering & Systems**: Google Bigquery, PostgreSQL, ETL/ELT Pipelines, Parallel Processing (Dask), Linux Server Maintenance
- **Dashboards & Visualization**: Plotly/Dash, Seaborn/Matplotlib, Web Applications (Flask/FastAPI)
- **Core Infrastructure:** Python, SQL, Git, Docker, CI/CD, AWS/GCP

Professional Experience

Data Scientist
Digital Science & Research Solutions, Ltd.
June 2022 - Present

- **LLM & Automated Annotation:** Constructed an LLM-backed document metadata and labeling pipeline to automatically fetch, categorize and summarize clusters across multiple document types for United States Federal Research Funding Agencies and Institutes.
- **Vector Databases & Scalable Search:** Conducted benchmarking across enterprise vector databases to optimize query latency and storage efficiency for **10M+ high-dimensional document vectors**.
- **ETL & Big Data Engineering:** Engineered automated ETL pipelines to extract external unstructured web/literature data alongside secure ingestion and processing of HHS-affiliated client's data using Python and SQL. Also, maintained and modified other ETL pipelines for processing **100GB+** of unstructured literature data.
- **Dashboards & Reporting:** Built interactive dashboards (Plotly/Streamlit) for internal use to explore dynamic topic clusters, research trends, and growth metrics in real time.

Graduate Researcher Scientist
University of Pennsylvania
August 2016 - June 2022

- **Large-Scale NLP & Computational Modeling:** Designed and deployed unsupervised NLP frameworks (**Word2Vec, FastText**) to analyze over 20,000 temporal semantic shifts across millions of unstructured scientific documents.
- **Pipeline Engineering & Big Data:** Built automated, parallelized Python pipelines to ingest, clean, and analyze **multi-gigabyte genomic and textual datasets**, reducing data processing bottlenecks by over 40%.

- **Statistical Modeling & Algorithm Optimization:** Applied high-dimensional dimensionality reduction (**UMAP**) and clustering algorithms (**K-Means**) to extract hidden patterns in complex biological networks.
- **Cross-Functional Collaboration & Technical Communication:** Collaborated with cross-departmental teams of bioinformaticians, software engineers, and domain experts; presented complex quantitative findings to non-technical stakeholders and external partners.
- **Software Development & Maintainability:** Maintained open-source codebase repositories on GitHub using strict version control (Git/GitHub), unit testing, and Docker containerization.

Publications

- **Unmasking The Language Of Science Through Textual Analyses On Biomedical Preprints And Published PapersOn Biomedical Preprints And Published Papers**
  Nicholson, D. N. (2022)
- **Changing Word Meanings in Biomedical Literature Reveal Pandemics and New Technologies**
  Nicholson, D. N. Alquaddoomi, F., Rubinetti, V., Greene, C. S. (2023)
- **Characterization of the Genome and Silk-gland Transcriptomes of Darwin's Bark Spider (Caerostris darwini)**
  Babb, P. L., Gregorič, M., Lahens, N. F.,**Nicholson, D. N.**, Hayashi, C. Y., Higgins, L., Kuntner, M., Agnarsson, I., Voight, B. F. (2022)
- **Examining Linguistic Shifts between Preprints and Publications**
  Nicholson, D. N., Rubinetti, V., Hu, D., Thielk, M., Hunter, L. E., Greene, C. S. (2022)
- **Expanding a Database-derived Biomedical Knowledge Graph via Multi- Relation Extraction from Biomedical Abstracts**
  Nicholson, D. N., Himmelstein, D. S., Greene, C. S. (2022)
- **Ten important roles for academic leaders to promote equity, diversity, and inclusion in data science.**
  Moore JH, Truong VQ, Robbins AB, **Nicholson D. N.**, Williams-Devane CL. (2021)
- **Constructing Knowledge Graphs and Their Biomedical Applications**
  Nicholson, D. N., Greene, C. S. (2020)
- **The Nephila Clavipes Genome Highlights the Diversity of Spider Silk Genes and their Complex Expression**
  Babb, P. L., Lahens, N. F., Correa-Garhwal, S. M., **Nicholson, D. N.**, Kim, E.J., Hogenesch, J.B., Kuntner, M., Higgins, L., Hayashi, C. Y., Agnarsson, I., Voight, B.F. (2017)

Education

Doctor of Philosophy (Ph.D.), Genomics and Computational Biology; University of Pennsylvania (Philadelphia, PA)

Postbaccalaureate Program (Penn Prep); University of Pennsylvania (Philadelphia, PA)

Bachelor of Science, Computer Science; University of Maryland Baltimore County (Baltimore, MD)